

GRANITE FALLS, MN, USA

WMO: 722033

Lat: 44.753N

Lon: 95.556W

Elev: 319

StdP: 97.55

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -25.4	(c) -22.8	(d) -28.8	(e) 0.3	(f) -25.1	(g) -26.3	(h) 0.4	(i) -22.7	(j) 14.4	(k) -11.3	(l) 12.8	(m) -9.8	(n) 3.1	(o) 260	(p) 0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	32.4	23.3	30.7	22.4	29.1	21.4	25.3	30.4	24.1	28.6	23.0	27.3	5.5	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	19.3	28.7	22.6	18.0	27.4	21.5	16.8	26.1	79.2	30.5	74.1	28.9	69.7	27.3	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.2	10.9	9.3	-30.3	35.4	3.2	1.3	-32.6	36.4	-34.5	37.1	-36.3	37.9	-38.6	38.8
			-30.4	27.3	3.1	1.1	-32.7	28.0	-34.6	28.7	-36.3	29.3	-38.6	30.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-9.5	-8.4	-0.2	7.3	14.3	20.4	22.4	20.6	16.8	8.7	1.0	-7.2
	DBStd	12.66	7.63	7.05	7.17	5.79	5.08	3.44	3.09	3.02	4.62	5.23	6.17	6.84
	HDD10.0	2472	603	514	323	116	18	0	0	0	4	87	275	532
	HDD18.3	4389	862	747	574	333	145	17	5	13	82	301	520	791
	CDD10.0	1476	0	0	7	35	151	311	383	327	209	47	5	0
	CDD18.3	352	0	0	1	2	20	78	130	82	37	3	0	0
	CDH23.3	3501	0	0	3	29	279	772	1298	699	373	47	1	0
	CDH26.7	1086	0	0	1	5	101	249	434	180	106	11	0	0
Wind	WSAvg	4.2	4.6	4.7	4.7	5.1	5.1	4.0	3.1	2.9	3.6	4.2	4.5	4.3
Precipitation	PrecAvg	704	17	18	40	65	83	109	88	90	79	64	32	20
	PrecMax	1014	53	39	103	166	162	248	179	201	210	181	109	68
	PrecMin	544	0	4	6	15	22	48	17	14	13	10	1	1
	PrecStd	139	14	11	24	35	39	52	41	43	47	42	32	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.1	12.5	22.0	26.8	33.5	34.3	34.1	32.3	32.4	28.0	20.1	9.8
		MCWB	3.9	9.9	13.9	15.6	18.3	22.7	25.9	24.9	22.4	16.9	12.5	6.6
	2%	DB	3.6	6.8	17.1	22.9	29.4	31.5	32.3	30.4	29.7	23.8	16.1	5.7
		MCWB	1.9	4.3	12.4	13.7	17.5	21.6	24.8	23.6	21.2	15.5	10.5	3.1
	5%	DB	2.2	3.4	12.8	20.0	26.6	29.7	30.6	28.7	27.6	20.3	12.5	3.0
		MCWB	1.0	1.7	9.9	11.8	16.6	20.9	23.4	22.5	20.0	13.9	8.1	1.7
	10%	DB	0.5	1.3	9.5	17.2	23.4	27.6	29.0	27.3	25.3	17.5	9.6	1.4
		MCWB	-0.2	0.2	6.9	10.7	15.2	19.5	22.2	21.1	18.8	12.1	6.3	0.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.3	9.9	15.7	17.3	21.4	25.3	27.5	26.0	24.6	19.4	14.3	7.3
		MCDB	7.1	12.7	19.4	22.9	28.5	31.4	32.5	30.7	29.2	24.3	19.3	9.0
	2%	WB	2.1	4.4	12.7	15.3	19.4	23.7	25.8	24.7	22.9	16.7	11.0	3.4
		MCDB	3.4	6.4	15.7	21.4	25.8	29.1	30.7	29.2	26.9	21.7	14.9	5.2
	5%	WB	1.1	2.0	9.9	13.3	17.9	22.1	24.5	23.4	21.6	14.7	8.7	1.9
		MCDB	2.0	3.3	13.5	17.9	24.2	27.5	29.3	27.4	26.0	19.4	12.1	3.1
	10%	WB	-0.1	0.3	7.0	11.5	16.5	20.8	23.3	22.3	20.1	12.5	6.5	0.6
		MCDB	0.5	1.3	9.4	16.3	21.7	25.9	27.8	26.0	24.0	16.5	9.7	1.4
Mean Daily Temperature Range	5% DB	MDBR	9.5	10.1	9.5	11.3	12.1	11.2	11.9	11.6	12.7	11.6	10.2	8.9
		MCDBR	10.4	11.5	14.6	17.2	16.9	13.9	12.5	12.9	14.9	16.1	15.2	10.1
		MCWBR	8.8	9.1	9.6	9.6	7.6	6.2	6.5	6.8	7.3	9.0	9.7	8.4
	5% WB	MCDBR	9.6	11.1	14.0	13.7	13.4	11.8	11.6	11.4	12.3	13.9	14.0	9.8
		MCWBR	8.1	8.9	9.4	8.5	7.1	6.4	6.7	6.6	7.1	8.6	9.4	8.3
Clear-Sky Solar Irradiance	taub		0.262	0.275	0.307	0.360	0.381	0.394	0.401	0.407	0.365	0.320	0.282	0.262
	taud		2.458	2.430	2.446	2.342	2.346	2.368	2.334	2.295	2.391	2.509	2.533	2.468
	Ebn at Noon		875	924	933	901	888	875	864	844	855	858	845	837
	Edh at Noon		69	85	97	118	121	119	122	123	102	79	64	62
All-Sky Solar Radiation	RadAvg		1.83	2.88	3.82	4.73	5.45	6.07	6.56	5.41	4.21	2.74	1.79	1.34
	RadStd		0.20	0.34	0.38	0.46	0.41	0.41	0.22	0.44	0.31	0.37	0.16	0.14

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (59 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page